

Engineering Memorandum

ENGINEERING MEMORANDUM (INTERN TECHNICAL ASSESSMENT)

TO: Technical Evaluation Committee
FROM: Eshwar T., AI/ML Engineering Intern Applicant
DATE: September 12, 2026
SUBJECT: End-to-End Technical Summary: Constrained Object Detection & Reasoning API (Civic Infrastructure Domain)

1. Domain & Dataset Sourcing Strategy

- **Domain Selected:** Civic Infrastructure Road Damage Domain, focusing specifically on non-COCO classes — **potholes, structural cracks, surface wear, and manholes.**
- **Justification**
 - Standard COCO pre-trained models are insufficient for production civic automation.
 - Municipal monitoring requires high-precision localization and classification of specialized, non-standard surface anomalies that generic datasets fail to represent.
- **Sourcing & Labeling**
 - The dataset was curated from public road-damage benchmarks.
 - It combines localized road surface captures, manual bounding box annotations, and multi-class labels covering the exact target taxonomy (potholes, structural cracks, surface wear, and manholes).
 - All classes utilized are strictly non-COCO, satisfying the core project constraint.

2. Train/Val/Test Split Strategy & Technical Justification

- **Split Ratio:** 70% Training, 15% Validation, 15% Test.
- **Justification**
 - Ensures sufficient data variance across diverse asphalt textures, weathering artifacts, and lighting conditions.
 - Reserves a clean, unseen hold-out test set to prevent data leakage.
- **Training Configuration**
 - Backbone: Ultralytics RT-DETR Large (`rt detr-l.pt`)
 - Epochs: 50
 - Batch size: 8
 - Image size: 640
 - Seed: 42
 - Environment: Kaggle GPU, guaranteeing full reproducibility.

3. Evaluation Methodology & Metrics Analysis

- **Metrics Tracked:** mAP@0.5, mAP@0.5:0.95, Precision, and Recall across all targeted classes.

- **What Metrics Tell Us**

- Self-reported metrics demonstrate strong bounding-box localization for high-contrast road defects like prominent potholes and linear cracks.
- A balanced precision-recall trade-off is maintained at standard IoU thresholds.

- **What Metrics Don't Tell Us**

- High aggregate mAP can mask class-specific ambiguities, such as geometric feature overlaps between circular manholes and surface depressions.
- Quantitative averages fail to expose these issues without rigorous qualitative error analysis.

4. Comprehensive Failure Case Analysis (The 5 Edge Cases)

- **Context:** Demonstrating production readiness requires a transparent understanding of model limitations. Below are five documented failure modes observed during evaluation, complete with root-cause analysis.

- **Class Confusion (Manholes vs. Potholes)**

- Circular metallic or concrete manhole covers occasionally trigger false-positive pothole classifications due to similar geometric contours, circular boundaries, and lighting gradients on asphalt surfaces.
- *Root Cause:* High visual and structural ambiguity between recessed manhole rims and deep potholes in grayscale intensity profiles.

- **Severe Occlusion**

- Partial blockage of road fractures and potholes by parked vehicles, debris, or pedestrians leads to missed detections or truncated bounding boxes.
- *Root Cause:* Lack of visible spatial context required by the feature pyramid network.

- **Motion Blur / Low-Shutter Artifacts**

- Images captured from fast-moving municipal vehicles exhibit lateral motion blur, washing out fine edges of structural cracks and surface wear.
- *Root Cause:* High-frequency spatial detail loss in the convolutional feature extractor.

- **Extreme Lighting & Glare**

- Direct noon sunlight reflecting off wet asphalt or metallic manhole surfaces creates high-intensity specular highlights, blinding the feature maps.
- *Root Cause:* Saturation clipping across pixel intensity channels.

- **Micro-Cracks & Scale Discrepancies**

- Hairline micro-cracks spanning large road segments fall below the effective receptive field at lower feature pyramid scales (`imgsz=640`).
- *Root Cause:* Resolution bottlenecks limiting pixel-level gradient propagation for tiny or thin objects.

5. Part B Reasoning Layer & Confidence Guardrail Logic

- **Architecture:** The natural-language reasoning layer (`/reason`) is implemented natively using the Groq API via standard `requests` , strictly avoiding bloated agent frameworks (such as LangChain or LangGraph), adhering to the constraints.

- **Intent Routing (Stage 1)**

- An initial classification prompt evaluates whether a user query requires visual context.
- Unrelated queries bypass the vision model entirely.

- **Inference vs. Guardrail Thresholds (Stages 2 & 3)**

- The vision model runs at an inference sensitivity threshold of `conf=0.25` to capture candidate regions for the target classes.
- A strict post-processing **Confidence Guardrail (0.35)** is enforced on top of this.

- **“Insufficient Information” Example**

- If an image contains ambiguous shadows or low-confidence detections below `0.35` (such as an unclear surface patch or blurred crack), the guardrail trips, logs a warning, and returns:
 - `{"status": "low_confidence", "response": "The detection results are insufficient to answer confidently. Human verification is required."}`
- This prevents the LLM from hallucinating answers over weak visual data.